

## **LSM3259 Creative Assignment Writeup**

By Joey Hoe

### **Stronger Together: The Ecological Importance of Mycorrhizal Relationships**

Ectomycorrhizal relationships are formed with both gymnosperms and angiosperm trees, evolving on several separate occasions from diverse saprotrophic lineages.<sup>1</sup> Ectomycorrhizal fungi form a protective mantle around the plant roots, known as a Hartig net. This protective mantle improves the tree's resistance to pathogens and stress. It also improves uptake of nutrients by the tree.<sup>2</sup> Ectomycorrhizal fungi are more adapted to harsh environments compared to endomycorrhizal fungi, helping plants survive extreme cold and poor nutrient availability. They can be intentionally introduced during ecosystem restoration, such as grass planting and afforestation, and improve the resilience of forest ecosystems to climate change.<sup>2</sup>

Arbuscular mycorrhizal fungi (AMF) are obligate symbiotic fungi that form relationships with approximately 80% of land plants. They belong to the phylum Glomeromycota.<sup>3</sup> They supply these land plants with inorganic nutrients, mainly phosphorus, but also nitrogen, trace elements and water.<sup>3</sup>

AMF is also capable of enhancing the phytoremediation capacity of plants, as they reduce the stress caused by pollutants.<sup>6</sup> AMF produces proteins that bind to heavy metals<sup>7</sup>, and promote the biodegradation of organic pollutants such as benzene.<sup>8</sup> The expansion of the mycelium also expands the rhizosphere influence zone, allowing the plant to access a larger volume of soil, including the nutrients and environmental pollutants contained within that soil.<sup>9</sup>

AMF receive up to 20% of photosynthetically fixed CO<sub>2</sub> as carbohydrates from the plant.<sup>4</sup> Plants transfer a total of 5 billion tons of carbon to arbuscular mycorrhizal fungi; thus they are important as a carbon sink.<sup>3,5</sup> The carbon sequestration of ectomycorrhizal fungi is even more efficient despite being less researched.<sup>2,14,15</sup> With a vital role in carbon storage, mycorrhizal fungi are a long-overlooked component of climate change mitigation.

Arbuscular mycorrhizal fungi differ from ectomycorrhizal fungi in that they grow into the plant root cells forming arbuscules, while ectomycorrhizal fungi grow on the outside of the root. In addition, many well-known mushroom species such as *Amanita muscaria* form ectomycorrhizal relationships with certain types of trees, such as pine and birch. The model depicts the underground structure of birch (*Betula* sp.) tree roots and the mycelium surrounding the roots, as well as the connecting paper model of *Amanita muscaria* (not to

scale) showing how the mycelium is connected to the fruiting body that is visible above ground.

Interestingly, not all plants form mycorrhizal relationships. Notably, Brassicaceae do not form mycorrhizal relationships for multiple reasons.<sup>10,11</sup> Firstly, mycorrhizal growth stimulator is absent in the roots of plants from the Brassicaceae family<sup>12</sup>. Secondly, many plants from this family produce anti-fungal compounds as a defence against pathogens, which has been hypothesized to be another reason for their non-mycorrhizal status.<sup>13</sup>

#### Citations:

1. F., Kohler, A., Murat, C. *et al.* Unearthing the roots of ectomycorrhizal symbioses. *Nat Rev Microbiol* **14**, 760–773 (2016). <https://doi.org/10.1038/nrmicro.2016.149>
2. Song W. Ectomycorrhizal fungi: potential guardians of terrestrial ecosystems. *mLife*. 2024;3:387–390. <https://doi.org/10.1002/mlf2.12127>
3. Schüßler, A., Martin, H., Cohen, D., Fitz, M., & Wipf, D. (2007). Arbuscular mycorrhiza: studies on the geosiphon symbiosis lead to the characterization of the first glomeromycotan sugar transporter. *Plant signaling & behavior*, 2(5), 431–434. <https://doi.org/10.4161/psb.2.5.4465>
4. Bago B, Pfeffer PE, Abubaker J, Jun J, Allen JW, Brouillette J, Douds DD, Lammers PJ, Shachar-Hill Y. Carbon export from arbuscular mycorrhizal roots involves the translocation of carbohydrate as well as lipid. *Plant Physiol*. 2003;131:1496–1507. doi: 10.1104/pp.102.007765.
5. Ouimette, A. P., Ollinger, S. V., Lepine, L. C., Stephens, R. B., Rowe, R. J., Vadeboncoeur, M. A., Tumber-Davila, S. J., & Hobbie, E. A. (2020). Accounting for Carbon Flux to Mycorrhizal Fungi May Resolve Discrepancies in Forest Carbon Budgets. *Ecosystems*, 23(4), 715–729. <https://www.jstor.org/stable/48730954>
6. Rabie GH (March 2005). "Role of arbuscular mycorrhizal fungi in phytoremediation of soil rhizosphere spiked with poly aromatic hydrocarbons". *Mycobiology*. **33** (1): 41–50.
7. Chen, Hansong; Xiong, Juan; Fang, Linchuan; Han, Fu; Zhao, Xiaolan; Fan, Qiaohui; Tan, Wenfeng (September 2022). "Sequestration of heavy metals in soil aggregates induced by glomalin-related soil protein: A five-year phytoremediation field study". *Journal of Hazardous Materials*. **437** 129445.
8. Fester T (January 2013). "Arbuscular mycorrhizal fungi in a wetland constructed for benzene-, methyl tert-butyl ether- and ammonia-contaminated groundwater bioremediation". *Microbial Biotechnology*. **6** (1): 80–4. [doi:10.1111/j.1751-7915.2012.00357.x](https://doi.org/10.1111/j.1751-7915.2012.00357.x).
9. Rajtor M, Piotrowska-Seget Z (November 2016). "Prospects for arbuscular mycorrhizal fungi (AMF) to assist in phytoremediation of soil hydrocarbon contaminants". *Chemosphere*. **162**: 105–16. [doi:10.1016/j.chemosphere.2016.07.071](https://doi.org/10.1016/j.chemosphere.2016.07.071)

10. Tester M., Smith S. E., Smith F. A. (1987). The phenomenon of "nonmycorrhizal" plants. *Can. J. Bot.* 65 (3), 419–431. doi: 10.1139/b87-051
11. Trautwig, A. N., Jackson, M. R., Kivlin, S. N., & Stinson, K. A. (2023). Reviewing ecological implications of mycorrhizal fungal interactions in the Brassicaceae. *Frontiers in plant science*, 14, 1269815. <https://doi.org/10.3389/fpls.2023.1269815>
12. Glenn M. G., Chew F. S., Williams P. H. (1988). Influence of glucosinolate content of Brassica (Cruciferae) roots on growth of vesicular–arbuscular mycorrhizal fungi. *New Phytol.* 110 (2), 217–225. doi: 10.1111/j.1469-8137.1988.tb00255.x
13. Anthony M. A., Celenza J. L., Armstrong A., Frey S. D. (2020). Indolic glucosinolate pathway provides resistance to mycorrhizal fungal colonization in a non-host Brassicaceae. *Ecosphere* 11 (4), e03100. doi: 10.1002/ecs2.3100
14. Liu B, Fan X, Meng D, Liu Z, Gao D, Chang Q, et al. Ectomycorrhizal trees rely on nitrogen resorption less than arbuscular mycorrhizal trees globally. *Ecol Lett.* 2024;27:e14346.
15. Terrer C, Vicca S, Hungate BA, Phillips RP, Prentice IC. Mycorrhizal association as a primary control of the CO<sub>2</sub> fertilization effect. *Science*. 2016;353:72–74.

Design choice for trees:

Birch and pine trees form ectomycorrhizal relationships with *Amanita muscaria*, which I chose because of its vibrant colours.